The Autonomous Scribe: Offloading Cognitive Burden in Fireground Command via LLM-Driven Task Tracking

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Abstract

In the chaotic initial minutes of a fire operation, the cognitive load on an Incident Commander (IC) often exceeds human capacity, transforming the radio into a bottleneck where critical tasks can be lost in the noise. This paper introduces an autonomous "digital scribe"—a hybrid AI system designed to listen to radio traffic, parse tactical intent, and visualize the operational picture in real-time. By integrating Large Language Models with human-in-the-loop verification, we explore whether AI can effectively serve as a cognitive backstop for emergency decision-making. Our evaluation reveals a distinct contrast between the system's high proficiency in identifying emerging assignments and the nuanced challenges of tracking their completion. We present the system architecture, scenario-based evaluation results, and feedback from active firefighters who identified an unexpected critical utility: the potential of AI to act as a temporal safety net against the silence of the fireground.

1 Introduction

Effective command and control on the fireground is predicated on a single, fragile variable: the ability of the IC to maintain situational awareness amidst chaos. In the initial minutes of a complex operation, an IC is subjected to a deluge of sensory inputs and radio traffic, often exceeding the cognitive capacity of a single individual. Research into the physiological stress response of firefighters demonstrates that the auditory alarm alone triggers a significant sympathetic nervous system surge, elevating heart rates significantly before operations even commence (Marciniak et al., 2021).

This state of hyper-arousal, while necessary for physical action, can impair working memory and the ability to track multiple parallel information streams, leading to increased error rates in cognitive tasks. The primary conduit for this information—the radio—remains a notorious bottleneck. While modern digital radio systems ensure signal clarity, they do not address the "fog of war" created by overlapping transmissions, unacknowledged orders, and the rapid decay of short-term memory under stress. A missed acknowledgment or a forgotten status update from an interior crew can lead to task saturation, where critical assignments are lost in the noise, potentially compromising firefighter safety and operational efficiency.

Furthermore, stress markers like epinephrine remain elevated for hours post-incident, suggesting that the biological toll of command extends well beyond the active fire fight (Horn et al., 2010).

Traditionally, the solution to this information overload has been the "command aide" or "scribe"—a dedicated human role to document tasks and benchmark progress. However, staffing constraints often leave the IC operating solo, forcing them to toggle between strategic decision-making and clerical record-keeping. This paper explores a technological solution to this gap: the integration of Artificial Intelligence (AI) and Natural Language Processing (NLP) into the command cycle.

We present the results of a novel AI-driven system designed to function as an autonomous "digital scribe" for emergency operations. Unlike passive recording devices, this system actively monitors radio frequency traffic, utilizing intent detection algorithms to identify specific tactical assignments (e.g., "Search and Rescue," "Ventilation") and track their lifecycle through to reported completion. By automatically parsing voice data into structured status boards and flagging uncompleted tasks or open communication loops, the system aims to offload the cognitive burden of memory from the IC. This paper evaluates the system's accuracy in task extraction and its potential to enhance closed-loop communication in high-stakes environments.

2 Related Work

Situational awareness is the cornerstone of effective command and control, yet it remains fragile under the combined pressure of time constraints, incomplete information, and physiological stress responses (Hollands et al., 2019). Research demonstrates that rapid message presentation rates and visual information displays substantially increase cognitive load and degrade situational awareness in command personnel, compared to slower rates and auditory presentation (Hollands et al., 2019).

Cognitive load and decision-making are intimately connected; high cognitive load drives personnel toward reflexive decision-making, while high task complexity necessitates deliberate, analytical reasoning (van Doorn, 2020). Digitization of the battlespace introduces a paradox: while automatic presentation of positional data reduces manual processing effort, it simultaneously decreases the depth to which commanders mentally process information, reducing their confidence in the situational picture (McGuinness & Ebbage, 2000).

Acute psychological stress in emergency operations exerts a bidirectional influence on field performance: moderate stress enhances alertness and focus, but excessive stress impairs working memory, executive function, and decision accuracy (Almutairi et al., 2025). Clear communication is essential for translating command intent into action; the "echo principle"—where receivers repeat back critical message elements—is the most effective radio communication practice (Caughey, 2020). In medical contexts, Closed-Loop Communication (CLC), a three-step model adapted from military radio protocols, has proven effective at reducing preventable errors (Salik & Ashurst, 2023).

Hybrid AI approaches demonstrate superior reliability in critical operations. Cercola et al. found that combined "LLM + Optimization" solutions outperformed purely language model-based approaches for high-stakes applications (Cercola et al., 2025). Speech recognition technology is critical for voice-based command systems in noisy environments; Moser et al. found that the commercial recapp system consistently outperformed open-source alternatives in Emergency Medical Services settings (Moser et al., 2025). The robustness of transcription directly impacts downstream language understanding; transcription error type influences accuracy, with corrections to named entities and nouns yielding the highest improvements (Shapira et al., 2025).

AI systems must present information in formats that enhance situational awareness rather than degrade it. Rastogi et al. found that analysts favor concise, context-rich, and role-specific explanations for AI-generated alerts, as overly complex explanations increase cognitive load and undermine trust (Rastogi et al., 2025). Objective measures of physiological load are essential for validating command support systems; Dias et al. identified Relative Heart Rate (%HRR) as the most frequently used metric for assessing cardiovascular load in occupational settings (Dias et al., 2023).

Existing research demonstrates that cognitive overload, stress-induced performance degradation, and communication failures represent critical vulnerabilities in command and control. While individual solutions have been developed and validated in isolation, few studies have integrated these findings into a cohesive AI-driven system designed for emergency command operations. This paper addresses that gap by presenting an autonomous digital scribe that combines speech recognition, natural language processing, closed-loop communication verification, and structured task tracking to reduce cognitive burden, enhance situational awareness, and ensure communication completeness on the fireground.

3 Use Case

The operational workflow, illustrated in Figure 1, is designed to support the command and control functions of the Feuerwehr Zug through a three-stage process: Input, Processing, and Output.

The use case initiates in the Input phase by aggregating heterogeneous data to synthesize a comprehensive operational context. Static knowledge bases—specifically the *Basic Regulations of Procedures* provided by the Swiss Fire Service Coordination and *Data sheets for specific Objects*—establish the foundational safety protocols and incident-specific measures. These resources are supplemented by the *Building Register*, which retrieves structural specifications from the openly accessible Swiss Building and Dwelling Register. Simultaneously, the system ingests a dynamic stream of radio communication transcripts, serving as the real-time input for the operation.

In the subsequent Processing phase, the system synthesizes these inputs to maintain a dynamic operational picture. As depicted in the central column of Figure 1, a *Baseline Tasklist* is generated based on the declared operation type and immediately augmented with object-specific tasks and relevant building information. This static framework is continuously updated by the *Ongoing Task Creation and Completion* module. By parsing transcribed radio traffic, the system identifies emergent tactical assignments and status updates, effectively merging pre-planned institutional knowledge with the fluid developments of the emergency ground.

The workflow culminates in the Output phase, where consolidated data is presented via the *Tasklist Dashboard*. This interface is designed as a decision-support tool rather than a fully autonomous agent; it incorporates a critical human-in-the-loop mechanism. As shown in the rightmost section of the figure, the Incident Commander reviews the automated entries, retaining the authority to *Confirm or Reopen* tasks. This validation step ensures that the automated scribe function remains aligned with expert human judgment before tasks are finalized, thereby reducing cognitive load while preserving ultimate operational control.

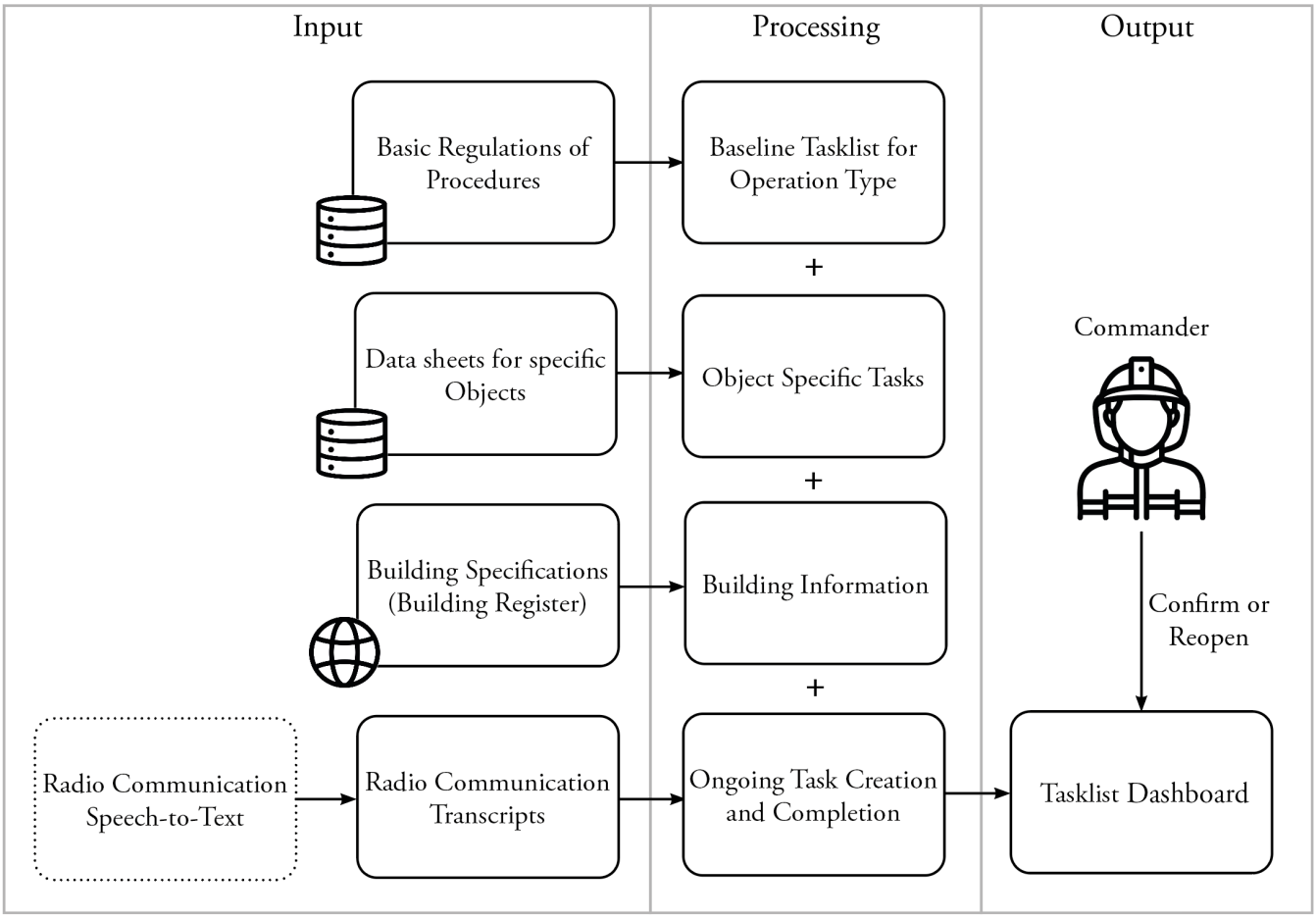

Figure 1: Workflow for AI-assisted task management during emergency operations.

4 Approach

This section describes the technical and conceptual design of the proposed AI-enhanced command support system. The approach follows a layered architecture that separates real-time communication, AI-based analysis, and human decision authority. This separation enables low-latency collaboration while ensuring robustness, interpretability, and human oversight in a high-stakes operational domain. The system is designed as a semi-autonomous assistant that augments, rather than replaces, human command decision-making. AI components perform continuous pattern recognition and knowledge retrieval, while final authority remains with the human operator.

4.1 System Architecture

The system implements a three-tier architecture comprising a Java Spring Boot backend, React frontend, and hybrid database layer. The backend provides RESTful API endpoints with WebSocket support for real-time bidirectional communication. The frontend employs component-based architecture with real-time state synchronization via WebSocket connections.

The database layer combines MariaDB for transactional data (sessions, messages, feedback) with PostgreSQL and pgvector extension for vector embeddings supporting semantic knowledge retrieval. This dual-database strategy balances ACID compliance for operational data with efficient semantic search capabilities.

4.2 Real-Time Communication Design

Traditional request-responseatterns introduce unacceptable latency in emergency communication scenarios. The system implements WebSocket connections for full-duplex communication, enabling rapid message delivery and acknowledgment updates across all connected participants. Each session maintains a dedicated WebSocket channel, eliminating polling overhead and ensuring synchronized situational awareness

mirroring operational radio communication. The prototype includes demonstration scenario scripts to simulate realistic radio traffic patterns.

Messages are structured as a self-referencing linked list, where each message contains a reference to its predecessor. This design enables conversation reconstruction after network disconnections, supports context-aware AI analysis by providing full conversation history, and guarantees chronological ordering independent of timestamp synchronization issues.

4.3 Communication Session Management

The system manages communication sessions with real-time message exchange between participants. The prototype implementation includes demonstration scenario scripts that simulate realistic radio traffic patterns, enabling system validation and feature demonstration. The system automatically parses message headers to extract sender, receiver, and directional metadata, supporting structured analysis of communication flows as shown in Figure 2.

4.4 AI-Enhanced Communication Analysis

4.4.1 Message Quality Assessment

The system leverages large language models to evaluate radio communications across three dimensions: clarity, completeness, and protocol adherence. AI analysis receives comprehensive context including Swiss-German firefighter radio protocols, message content, and session history, enabling context-dependent evaluation that significantly reduces false positives compared to isolated message analysis.

Feedback is categorized into five types: unclear formulation, logical errors, ambiguity, missing acknowledgments, and protocol violations. Each feedback item includes severity level, affected text excerpt, and specific improvement suggestions. Analysis runs asynchronously to prevent blocking the dispatch workflow, with status tracking enabling responsive UI interactions without timeout errors.

4.4.2 Holistic Acknowledgment Detection

Swiss fire service protocols mandate closed-loop communication where every command must be explicitly acknowledged. The system implements automatic acknowledgment detection using AI-powered intent classification with a critical design evolution:

Initial single-message approach checked only the immediately preceding message, which failed when multiple messages were dispatched before acknowledgment or when acknowledgments referenced earlier commands superseded by recent transmissions.

Revised holistic approach analyzes the complete message history collectively, instructing the AI to identify all confirmation pairs in the conversation. This enables detection of delayed acknowledgments, implicit confirmations (e.g., completion reports confirming earlier commands), and resolution of ambiguous acknowledgments when multiple commands are pending.

4.4.3 Human-in-the-Loop Verification

Fully automated AI systems introduce risks in high-stakes domains where false positives could lead to missed tasks. We implemented a three-state confirmation system:

1. **Unconfirmed**: Messages requiring acknowledgment but not yet evaluated
2. **AI-Confirmed**: AI suggests acknowledgment, awaiting human verification
3. **User-Verified**: Human operator has confirmed or denied AI suggestion

The system displays color-coded badges indicating acknowledgment status, with persistent controls allowing operators to change decisions as situational context evolves. Once a user makes a decision, the AI will not override that determination, respecting human authority as final decision-maker. This design creates a semi-autonomous system where AI provides intelligent assistance while ensuring human control over critical decisions, balancing automation efficiency with human judgment and reducing cognitive load while maintaining situational awareness.

4.5 Knowledge Management and Retrieval

4.5.1 Retrieval-Augmented Knowledge Access

Emergency responders require immediate access to operational knowledge: building layouts, hazardous material inventories, and specialized procedures. Traditional keyword-based retrieval fails when queries use synonyms or conceptually related terms. We implemented semantic search using vector embeddings where documents are converted into high-dimensional vector representations capturing semantic meaning.

Documents are split into chunks with overlap, embedded using the local `multilingual-e5-large` model (via ONNX Runtime), and stored in PostgreSQL with pgvector extension for efficient similarity search. Unlike cloud-based embedding services, this approach processes sensitive operational data entirely within the secure network perimeter. Query processing embeds user queries locally, retrieves top-K most similar chunks using cosine similarity, and augments LLM responses with retrieved context. Metadata filtering enables domain-specific searches (e.g., restricting to hazmat datasheets during chemical incidents).

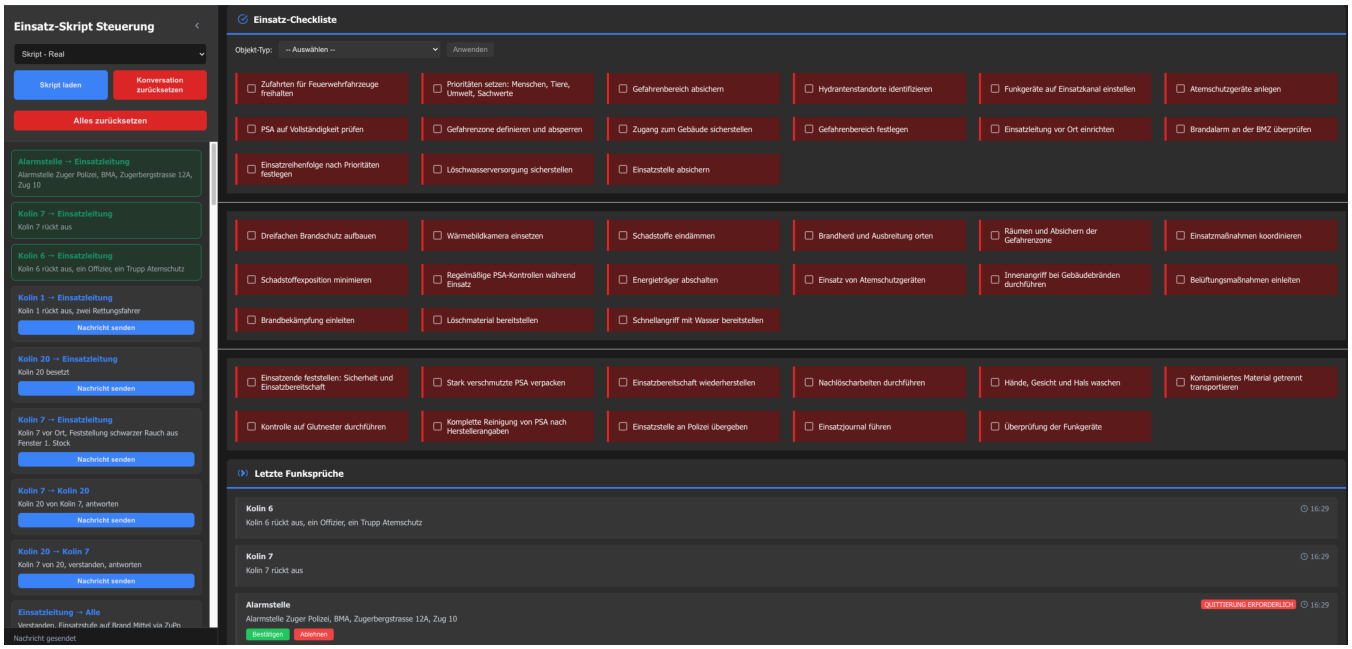

Figure 2: Emergency Session Dashboard with Real-Time Message Stream, AI Feedback, and Tasklist. Left Panel: Control to send mocked radio messages simulating emergency communication.

4.5.2 Geospatial Building Information Integration

The system integrates with Swiss federal databases to provide location-based emergency intelligence. The Swiss GeoAdmin API geocodes addresses to GPS coordinates, while the Federal Building and Dwelling Register retrieves building properties including construction year, number of floors, approximate height, occupancy estimates, and building category as indicated in Figure 3.

When an address is entered, the system geocodes the location, queries building properties, and retrieves building-specific operational documents from the RAG system using geospatial proximity search. This integration provides Incident Commanders with critical information within seconds of dispatch, without manual document lookup during high-stress initial response phases.

4.5.3 Automated Checklist Generation

The system provides automated checklist generation from PDF operational manuals using AI-powered document analysis. When operational procedures are uploaded, the system extracts text content using Apache PDFBox with fallback to Spring AI PDF readers, then processes the document through a multi-stage AI pipeline. The uploaded documents are stored in the relational database as shown in Figure 4.

The AI analysis focuses exclusively on operational steps that responders execute on-scene, filtering out administrative content, copyright information, and documentation metadata. Large documents are automatically split into manageable chunks to handle API token limits, with each chunk analyzed independently and results merged. The AI identifies actionable steps, generates concise checklist items (maximum 50 characters), and assigns operational phases (INITIAL, OPERATION, CLOSING) based on typical incident progression.

A secondary AI pass orders the extracted items chronologically according to standard fire service procedures and refines phase assignments. This two-stage approach ensures checklists reflect realistic tactical sequences. The system deduplicates items across chunks and summarizes if more than 25 items are generated, prioritizing the most critical operational steps. Generated checklists are immediately broadcast to connected clients via WebSocket, providing real-time availability without page refresh.

4.5.4 Dynamic Task Extraction and Tracking

Beyond baseline checklists from operational documents, the system architecture supports dynamic task extraction from radio communications in real-time. The AI analyzes message content to identify commands, requests, and assignments, creating corresponding task entries that supplement pre-defined checklists. By automating task tracking, the system reduces cognitive load on Incident Commanders who would otherwise need to manually record and monitor assignments while simultaneously managing dynamic emergency situations.

Task completion can be tracked through explicit status

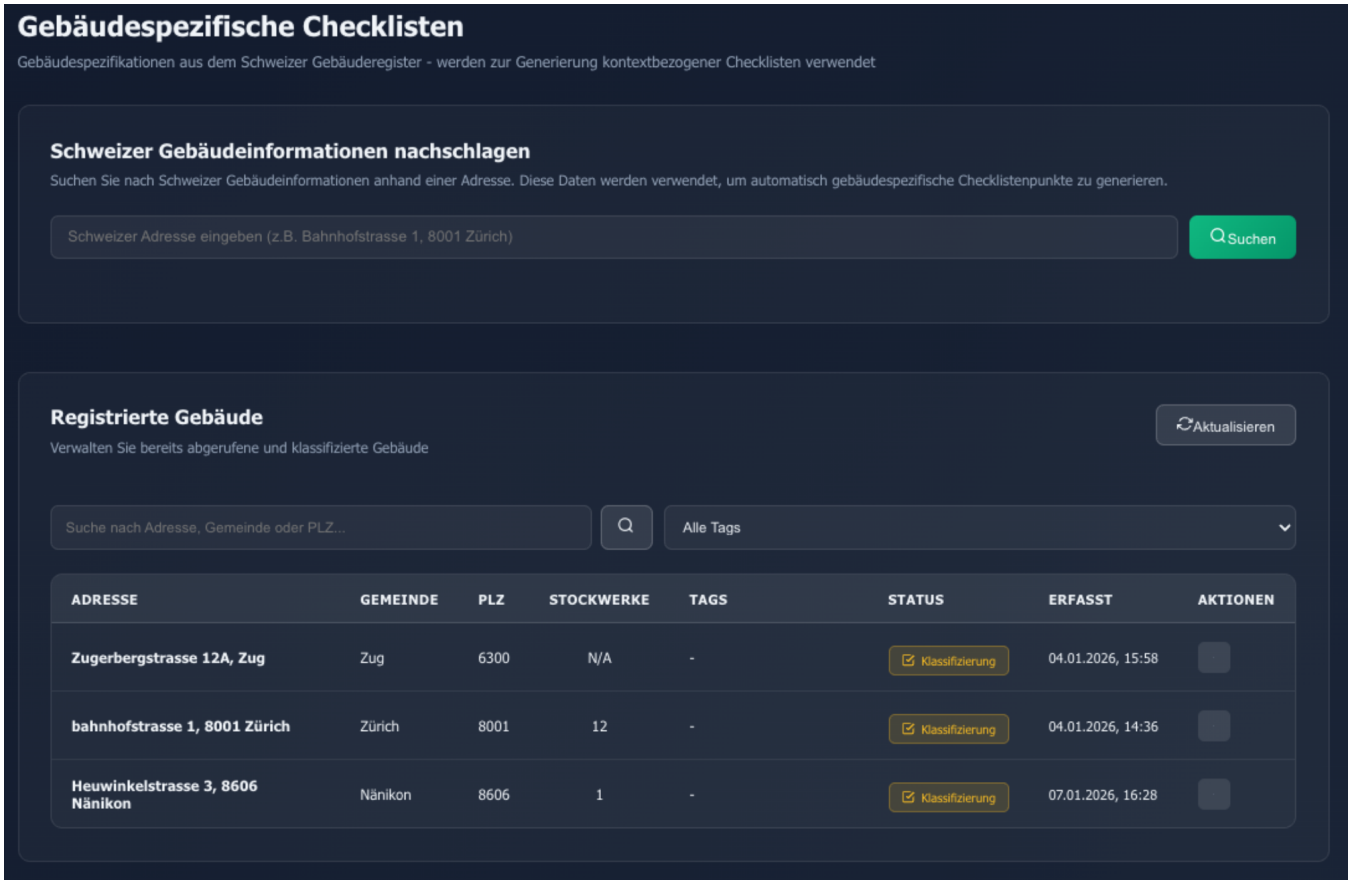

Figure 3: List of retrieved building information with the option to upload additional documents.

reports or inferred from subsequent communications indicating task fulfillment. This bidirectional task management—combining static baseline requirements with dynamic emergent tasks—provides comprehensive assignment tracking throughout incident progression.

4.6 Design Decisions and Alternatives

Stateless AI Architecture: We adopted stateless AI calls passing full conversation context with each request, rejecting stateful agent approaches that introduced complexity in lifecycle management and risked state corruption. This design improves robustness and debuggability.

Server-Side Processing: Client-side AI processing was rejected due to insufficient model capability for nuanced communication analysis and privacy risks from exposing training data to clients.

Relational Database Choice: Despite considering NoSQL for flexibility, relational databases provide ACID transactions essential for consistency between messages, acknowledgments, and feedback, with efficient join operations for complex queries.

Local vs. Cloud Embeddings: We prioritized a local embedding strategy using `multilingual-e5-large` over cloud-based alternatives (e.g., OpenAI `text-embedding-ada-002`). While this reduces vector dimensionality (384 vs. 1536), it eliminates data privacy risks regarding sensitive operational documents and significantly reduces retrieval latency ($\sim$25ms local CPU vs. $\sim$150ms API). This decision represents a strategic step toward the edge-compute goals outlined in Section 7.

4.7 Implementation Challenges

Key challenges included multi-message acknowledgment ambiguity when determining which acknowledgment corresponded to which command, addressed by providing detailed conversation context with message IDs. WebSocket connection stability was ensured through automatic reconnection with exponential backoff and full message history resynchronization. AI prompt engineering required iteration to reliably produce structured JSON output. Real-time performance was achieved by moving AI analysis to asynchronous workers with loading indicators. Swiss-German terminology support

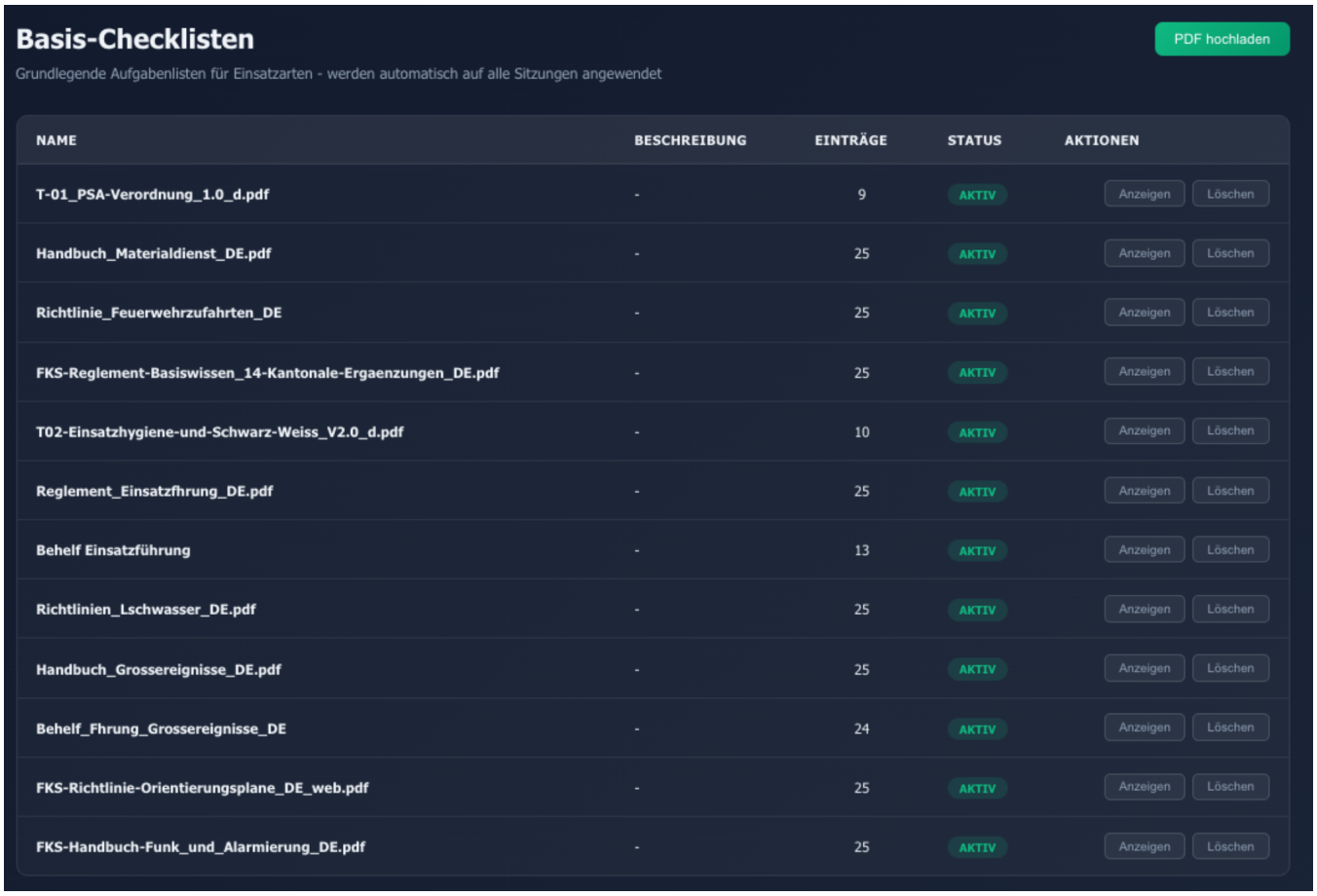

Figure 4: Checklist management interface showing generated checklists from operational documents.

was enabled through fire service glossaries in system prompts.

5 Evaluation

As this research represents an initial exploration into AI-assisted command rather than a deployment-ready product, evaluation focused on functional viability and architectural constraints. We assessed the system using a combination of synthetic scenarios and real-world transcripts provided by the Feuerwehr Zug.

5.1 Methodology

The system was tested against three distinct operational scenarios: two synthetic training sets (a standard single-family house fire and a multi-vehicle accident) and one anonymized transcript from a real structural fire incident. Each scenario contained between 20 and 40 distinct message exchanges.

To isolate the performance of the NLU (Natural Language Understanding) components from transcription errors, text was injected into the system via the WebSocket interface at a pace mimicking natural conversation flow, as described in Section 4.2.

5.2 Task Extraction and State Tracking

The system demonstrated a strong capability in **Task Extraction**, correctly identifying approximately 80% of explicit tactical commands (e.g., "Establish water supply," "Search 2nd floor"). The intent detection algorithms successfully distinguished between informative updates and actionable orders.

However, **Task Completion Tracking** proved to be the primary challenge. While explicit closures (e.g., "Task complete") were handled correctly, the system struggled with implicit closures—situations where a crew's report (e.g., "We are exiting the building") implies the completion of a previous order (e.g., "Search the interior"). This confirms that simple intent detection is insufficient for state management; a broader context window or "Chain of Thought" reasoning is required to resolve these implicit dependencies.

5.3 System Latency

In the current prototype configuration, despite the optimization of local retrieval mechanisms ($\sim$25ms), the end-to-end latency—from message receipt to dashboard update—averaged approximately 5 seconds. This delay is primarily attributed to the synchronous chaining of multiple cloud-based LLM calls (Intent Detection $\rightarrow$ Entity Extraction $\rightarrow$ JSON Formatting) required for generative reasoning. While acceptable for a proof-of-concept, field deployment would require optimization to achieve sub-second performance, likely through parallel processing or smaller, fine-tuned models running at the edge.

5.4 Document Analysis

The automated checklist generation (PDF parsing) performed reliably across standard operational documents. The system successfully segmented long manuals into logical, phase-specific checklists (e.g., "Arrival," "Deployment," "Termination") without significant hallucinations, validating the RAG approach for static knowledge retrieval.

5.5 Qualitative User Feedback

We conducted a demonstration session with active members of the Feuerwehr Zug. Feedback was predominantly positive regarding the concept of a "Digital Scribe." Notably, the responders emphasized that the system's value lay not just in recording text, but in **temporal tracking**. The ability to see exactly *how long* a task has been "Open" or how long a crew has been silent was identified as a critical safety feature—a "temporal safety net" that human scribes often struggle to maintain during high-stress saturation events.

6 Conclusion

This study presented the design and evaluation of an AI-driven digital scribe aimed at mitigating cognitive overload for Incident Commanders. By integrating semantic search, intent detection, and automated task management, the system seeks to transition the command workflow from a reliance on fragile individual memory to a supported human-machine teaming model.

Evaluation of the prototype yields a mixed but promising functional picture. The system demonstrated high reliability in the extraction of tactical assignments from communication streams, successfully identifying and cataloging tasks such as search operations or ventilation orders as they occurred. However, the automated lifecycle tracking—specifically the ability to algorithmically determine when a task has been completed based on implicit dialogue cues—proved less robust. While the creation of the "operational picture" is reliable, the maintenance of its state currently requires higher levels of human intervention than anticipated.

Qualitative validation with active firefighters highlighted the distinct value of temporal tracking. Participants noted that the system's ability to monitor the duration of open tasks provides a critical safety net, alerting commanders to operations that may have stagnated or crews that have gone silent. Despite current latency limitations inherent to the prototype configuration, the feedback confirms that offloading the memory burden allows the IC to focus on strategic decision-making rather than clerical record-keeping. Ultimately, this work establishes that while AI cannot replace the intuition of a commander, it can effectively serve as a resilient informational backstop in high-stress environments.

7 Future Research

While the current prototype validates the architectural approach, several critical avenues for research and development remain to bridge the gap between simulation and field deployment.

7.1 Robust Speech-to-Text in Acoustic Extremes

The current study relied on pre-transcribed messages to isolate the performance of the NLU (Natural Language Understanding) components. However, the real-world utility of this system is entirely dependent on the quality of the input stream. Future research must focus on integrating and evaluating Speech-to-Text (STT) engines specifically trained on low-fidelity, high-noise radio channels. This includes handling the "Lombard effect" (speech changes under stress), background mechanical noise (pumps, saws), and specific dialectal variances (e.g., Swiss German) common in local brigades.

7.2 Context-Aware State Tracking

To address the observed limitations in task completion tracking, future iterations should move beyond single-message analysis. Developing a state-tracking model that utilizes a broader temporal context window could help the system understand implicit completion (e.g., a crew reporting "Ready for assignment" implies their previous task is done). Research into "Chain-of-Thought" prompting or specialized fine-tuned models may improve the accuracy of these state transitions.

7.3 Edge AI and Offline Resilience

The current architecture relies on cloud-based LLM APIs for the generative reasoning step, introducing latency and requiring consistent internet connectivity—a luxury not guaranteed at disaster sites. While the retrieval system (RAG) has been

successfully migrated to local execution using ONNX models, the generation phase remains a bottleneck. Future work will investigate the distillation of large models into smaller, specialized models capable of running locally on command vehicle hardware (Edge AI). This would eliminate network latency and ensure the system remains functional regardless of cellular infrastructure status.

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